

HW8: Holt's Trend-Corrected Exponential Smoothing

ISYE 4031 - Summer 2026 - Problems Only Version

Submission Instructions

Include your full name, Georgia Tech email, and GTID in your submitted work. You may complete the homework using JupyterLite, a local Python or R environment, spreadsheet software, handwritten work where appropriate, or another approved environment. Your submission should clearly answer each question and show enough reasoning, formulas, code output, or supporting work to be graded.

Required Data Files

- calculator_sales_hw8.csv

Problem Data

For the past two years, a department store has carried a new type of electronic calculator. Sales have generally been increasing. The store wants enough calculators to meet practically all demand.

The data are in calculator_sales_hw8.csv.

Problem 1(a): Initial Level and Trend

Find the initial values l_0 and b_0 by fitting a straight line to the first half of the data, meaning the first 12 months.

Problem 1(b): Holt Recursion Table

Set up Holt's trend-corrected exponential smoothing recursion. Your table should include at least observed sales, one-step forecast, level, trend, error, and squared error.

Use the formulas:

$$l_t = \alpha y_t + (1 - \alpha)(l_{t-1} + b_{t-1}),$$

$$b_t = \gamma(l_t - l_{t-1}) + (1 - \gamma)b_{t-1},$$

$$\hat{y}_{t+1|t} = l_t + b_t.$$

Problem 1(c): Optimize Smoothing Constants

Find values of α and γ that minimize SSE subject to $0 \leq \alpha \leq 1$ and $0 \leq \gamma \leq 1$. Mention what tool or software you used.

Problem 1(d): Forecast Year 3 January, February, and March

Find point forecasts and approximate 95% prediction intervals for January, February, and March of year 3.

Problem 1(e): Short Report

Write a short report explaining your findings. Include the data pattern, initial values, optimized constants, forecast table, and interpretation of the prediction intervals.

Problem 1(f): Work File / Spreadsheet Equivalent

Submit a work file that lets the grader verify how the forecasts were produced. If you use JupyterLite, submit the executed notebook. If you use spreadsheet software, submit the spreadsheet file. If you use another environment, include the relevant code/output or work file.

AI Usage Report

Your submission must include an AI Usage Report following the course AI policy. If you used AI tools, include the tool(s), what you asked them to help with, what output you accepted, changed, or rejected, and how you verified the final work. If prompts or screenshots materially affected your submission, include copied prompt text or screenshots in the report when appropriate. If no AI tools were used, include this statement: "No AI tools were used for this submission. You may include this report at the end of your submission or attach it as a separate file.